Assignment -10

```
#include <stdio.h>

int main() {
    int choice;
    double num1, num2, result;
    int i, n;
    long fact = 1;

    printf("--- Simple Calculator ---\n");
    printf("1. Addition\n2. Subtraction\n3. Multiplication\n4. Division\n5. Power (x^y)\n6. Factorial
(x!)\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("Enter two numbers: ");
            scanf("%lf %lf", &num1, &num2);
            printf("Result = %.2f\n", num1 + num2);
            break;
        case 2:
            printf("Enter two numbers: ");
            scanf("%lf %lf", &num1, &num2);
            printf("Result = %.2f\n", num1 - num2);
            break;
        case 3:
            printf("Enter two numbers: ");
            scanf("%lf %lf", &num1, &num2);
            printf("Result = %.2f\n", num1 * num2);
            break;
        case 4:
            printf("Enter two numbers: ");
            scanf("%lf %lf", &num1, &num2);
            if(num2 != 0)
                printf("Result = %.2f\n", num1 / num2);
            else
                printf("Error: Division by zero!\n");
            break;
        case 5:
            printf("Enter base and exponent: ");
            scanf("%lf %lf", &num1, &num2);
            result = 1;
            for(i = 1; i <= num2; i++) {
                result *= num1;
            }
            printf("Result = %.2f\n", result);
            break;
        case 6:
            printf("Enter a number: ");
            scanf("%d", &n);
            for(i = 1; i <= n; i++) {
                fact *= i;
            }
            printf("Factorial = %ld\n", fact);
            break;
        default:
            printf("Invalid choice!\n");
    }
    return 0;
}
```

Assignment -10

main.c

```c
#include <stdio.h>

int main() {
    int choice;
    double num1, num2, result;
    int i, n;
    long fact = 1;

    printf("--- Simple Calculator ---\n");
    printf("1. Addition\n2. Subtraction\n3. Multiplication\n4. Division\n5. Power (x^y)\n6. Factorial (x!)\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1:
            printf("Enter two numbers: ");
            scanf("%lf %lf", &num1, &num2);
            printf("Result = %.2f\n", num1 + num2);
            break;
        case 2:
            printf("Enter two numbers: ");
            scanf("%lf %lf", &num1, &num2);
            printf("Result = %.2f\n", num1 - num2);
            break;
        case 3:
            printf("Enter two numbers: ");
            scanf("%lf %lf", &num1, &num2);
            printf("Result = %.2f\n", num1 * num2);
            break;
        case 4:
            printf("Enter two numbers: ");
```

Output

```
--- Simple Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial (x!)
Enter your choice: 1
Enter two numbers: 2 3
Result = 5.00


=== Code Execution Successful ===
```